PROBLEM 4 · PROBLEM STATEMENT

A brick column cooled by air on one side

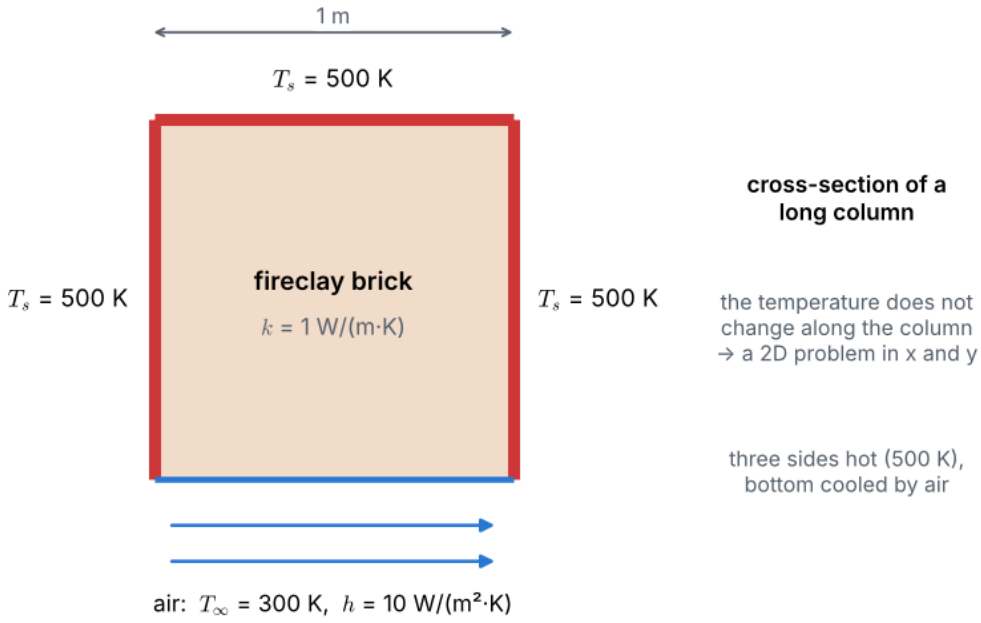
Two-dimensional steady conduction with a convection boundary, solved with the Finite Difference Method

The problem

A large industrial furnace is supported on a long column of **fireclay brick**, which is **1 m by 1 m** on a side. During steady-state operation, three surfaces of the column are maintained at **500 K**, while the remaining surface is exposed to an **air stream** for which $T_\infty = 300$ K and $h = 10$ W/(m²·K).

Using a grid of $\Delta x = \Delta y = 0.25$ m, determine the **two-dimensional temperature distribution** in the column and the **heat rate to the air stream per unit length** of column.

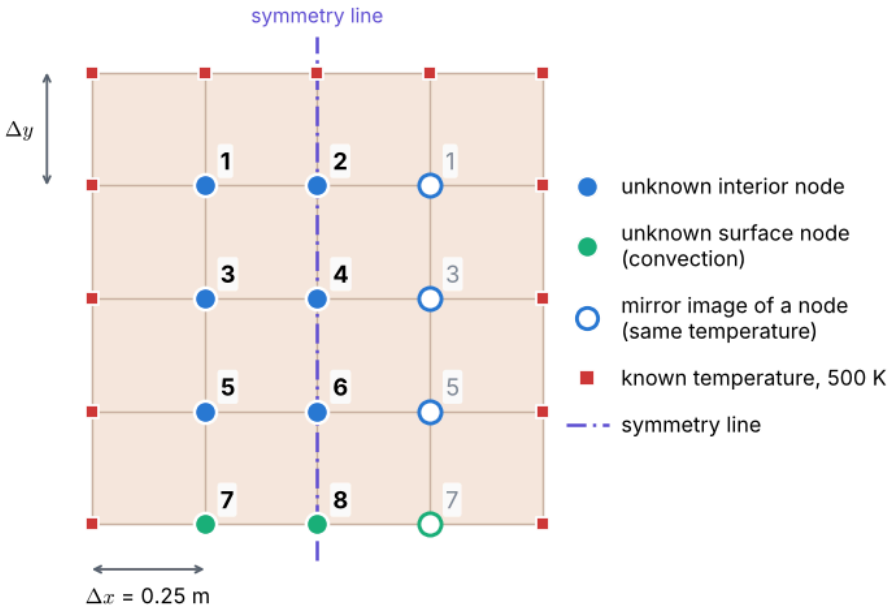
(Adapted from Incropera et al., Fundamentals of Heat and Mass Transfer, Example 4.3.)



Cross-section of the column: the top, left and right surfaces are held at 500 K; the bottom surface is cooled by the air stream.

Data

Quantity	Symbol	Value
width and height of the cross-section	W	1 m
grid spacing	$\Delta x = \Delta y$	0.25 m
thermal conductivity of fireclay brick (at about 478 K)	k	1 W/(m·K)
temperature of the top, left and right surfaces	T_s	500 K
air temperature	T_∞	300 K
convection coefficient at the bottom surface	h	10 W/(m²·K)



The grid with $\Delta x = \Delta y = 0.25$ m. Because of the symmetry line, only the eight numbered temperatures are unknown.

Assumptions

- Steady state.
- Two-dimensional conduction: the column is long and nothing changes along its length.
- No heat generation inside the brick.
- Constant thermal conductivity and convection coefficient.
- Radiation is neglected.

Tasks

1. **Physics.** Write the governing equation in the brick and the boundary condition on each surface. What is the Biot number $Bi = h\Delta x/k$ of a grid cell?

2. **Grid and symmetry.** Draw the grid. Which node temperatures are known and which are unknown? Use the symmetry of the problem to reduce the number of unknowns to eight, numbered 1 to 8 as in the grid figure above.
3. **Discretization.** Derive the node equations with the **energy balance method**: an interior node, a node on the convective surface (half control volume), and a node on the symmetry line.
4. **Solution.** Solve the eight equations by **Gaussian elimination** and by **Gauss–Seidel iteration**.
5. **Results.** Show the temperature distribution over the whole cross-section and calculate the **heat rate to the air** per unit length.
6. **Verification.** Check that the heat conducted in from the hot surfaces equals the heat convected out. Then solve the problem on finer grids: how do the temperatures and the heat rate change?

What you will learn

- How to derive node equations from **energy balances on control volumes**, including a **convection boundary**.
- How **symmetry** reduces the number of unknowns.
- How to calculate a **heat rate** from a discrete temperature field.
- Why a grid that is fine enough for temperatures may not be fine enough for **heat fluxes near corners**.

> **Files for this problem:** `02_solution.pdf` explains the solution step by step. `03_fdm_solver.ipynb` contains the complete Python code (plain Python, no numerical libraries), with explanations and plots, to run cell by cell.